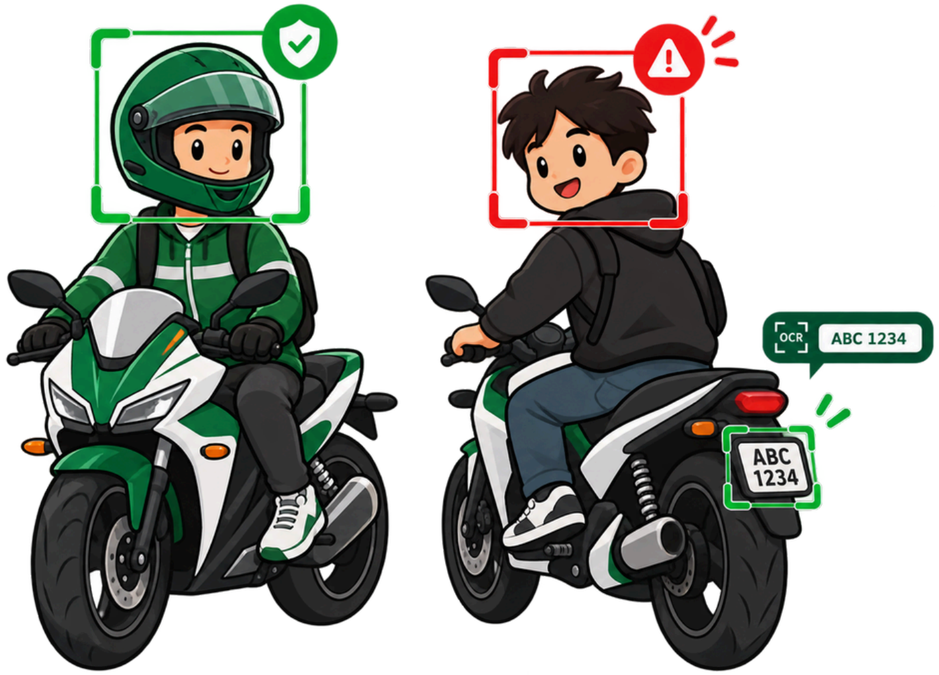
Motorcycle Helmet Violation Detection

Deep Learning-based traffic safety system using YOLOv8 + EasyOCR



Presenter: Lim Potkolbotey
Course: Deep Learning
Lecturer: Mr. Soklong HIM
School: Kirirom Institute of Technology
Date: 25 September 2026

Github: https://github.com/Jingjing468/motorcycle-helmet-violation-detection.git

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Project Overview

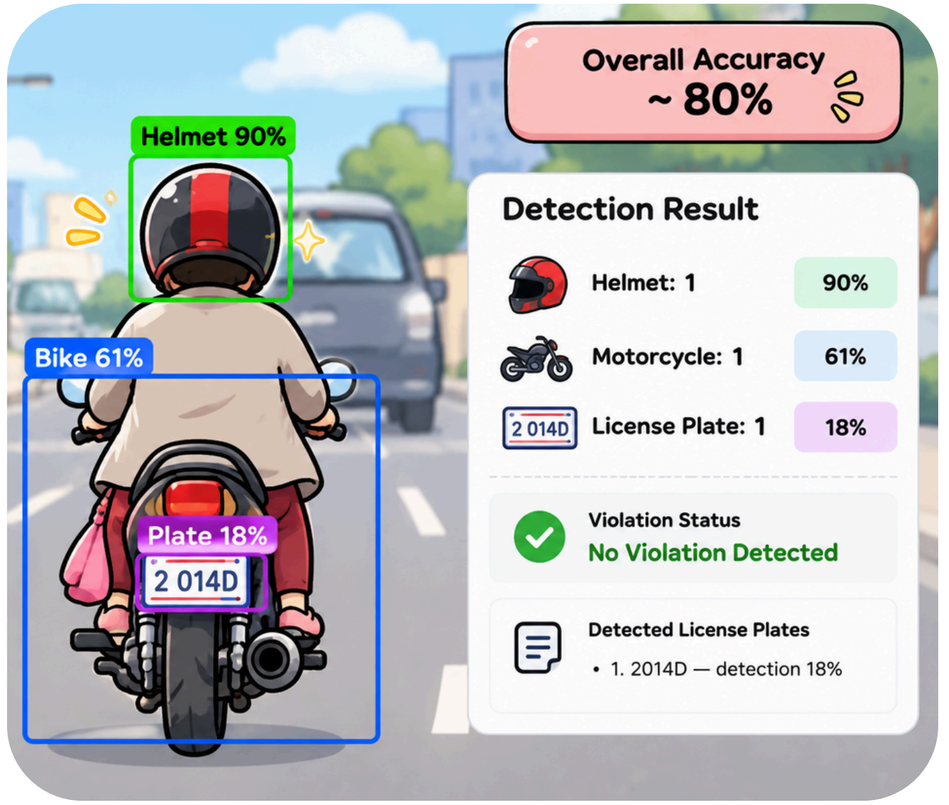
Main focus: helmet and no-helmet violation detection

Main Feature

- Helmet / No-Helmet Detection
- Detects helmet and no-helmet riders
- Identifies helmet violations
- Shows bounding boxes and confidence
- Runs in a Gradio web app

Supporting Features

- Motorcycle Detection
- Helps locate motorcycles and associate riders.
- License Plate Detection + OCR
- Detects plate regions and reads text using EasyOCR.



Problem Statement

Manual helmet monitoring is difficult in real traffic scenes

- Some riders do not wear helmets
- Crowded traffic is hard to monitor manually
- Small or blocked heads are difficult to detect
- Deep learning can support helmet monitoring



Project Scope

Clear focus helps explain the final system

PRIMARY OBJECTIVE

🪖 Helmet / No-Helmet Detection

✓ Detect helmet riders
✓ Detect no-helmet riders
✓ Identify helmet violations
✓ Evaluate detection performance

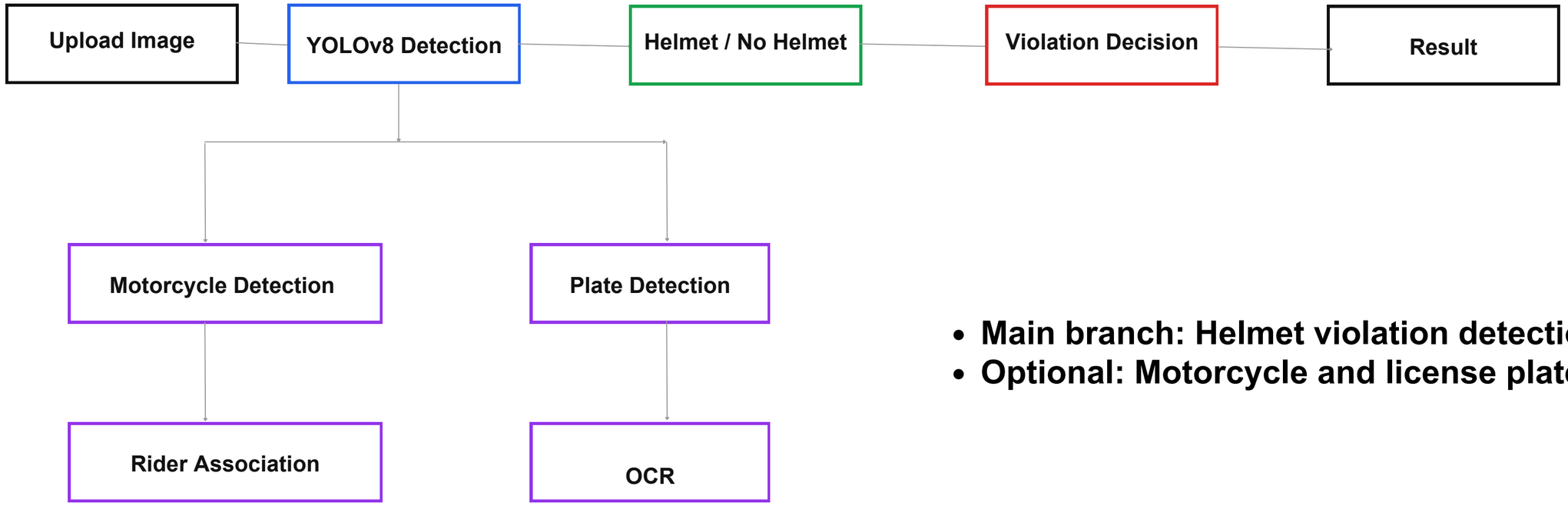
SUPPORTING FEATURES

🏍️ Motorcycle Detection

- Rider association
- 🔎 License Plate + OCR
- EasyOCR recognition
- FUTURE
- Improve these supporting features

System Workflow

From uploaded image to violation decision



- **Main branch: Helmet violation detection**
- **Optional: Motorcycle and license plate features**

Dataset & Pipeline



Dataset

8,481 images | Roboflow Version 4

Data Split

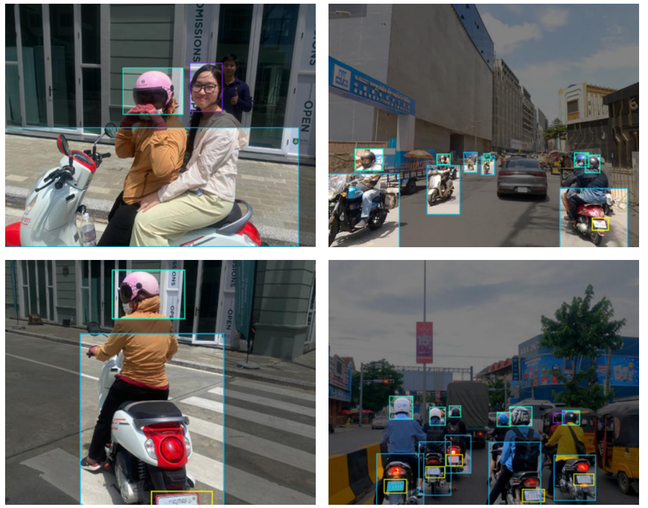
Train: 5,936 (70%)
Validation: 1,690 (20%)
Test: 855 (10%)

Classes

Bike — supporting
Helmet — main focus
No-Helmet — main focus
Number-Plate — optional

Images → Labeling → Split → Resize 640×640 → YOLOv8 Training

Dataset & Pipeline



Dataset Final Version 5

- 8,494 total images
- Train: 5,949
- Validation: 1,690
- Test: 855

New Local Data

- 13 Phnom Penh road images
- Koh Pich
- Toul Tompoung
- Manually labeled

Original Dataset → Local Phnom Penh Images → Labeling → Version 5 → Training

Local Phnom Penh Dataset





Koh Pich

- Road traffic images
- Different camera distances
- Real Cambodian motorcycles
- Helmet / No-Helmet cases



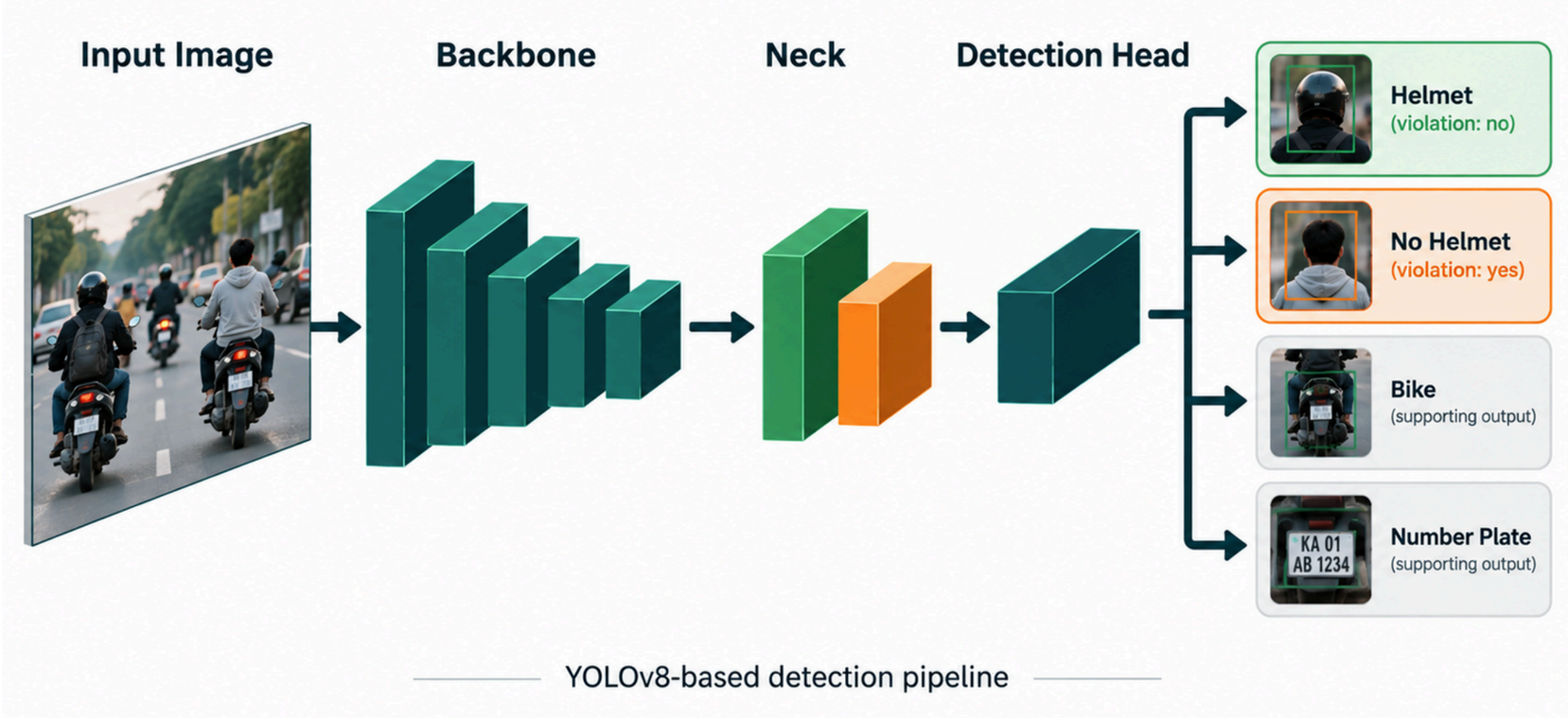


Toul Tompoung

- Crowded road scenes
- More motorcycles
- Occlusion and traffic
- Helmet / No-Helmet cases

Model Architecture & Training Strategy

YOLOv8 architecture, fine-tuning strategy, and tools



Training Losses

Box Loss • Classification Loss • DFL Loss

Model Architecture & Training Strategy

YOLOv8 architecture, fine-tuning strategy, and tools

TRAINING STRATEGY

Baseline Model
Original YOLOv8 trained on the dataset

Fine-Tuned Model
Continued training with difficult no-helmet examples

TOOLS

PyTorch + Ultralytics

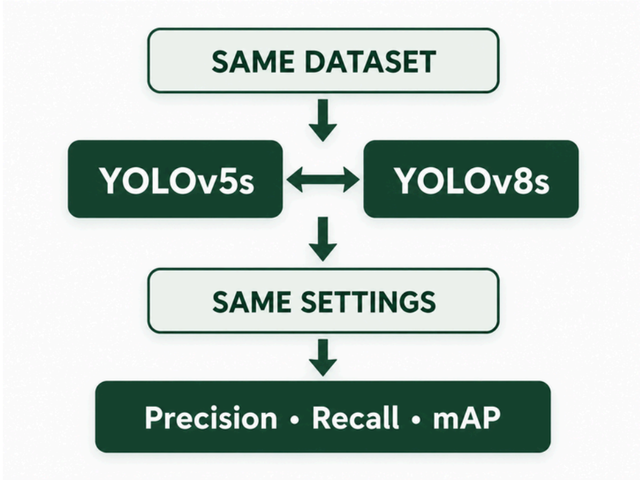
Google Colab + Tesla T4

Gradio + EasyOCR

Model Architecture Comparison

YOLOv5s vs YOLOv8s on the same Version 5 dataset

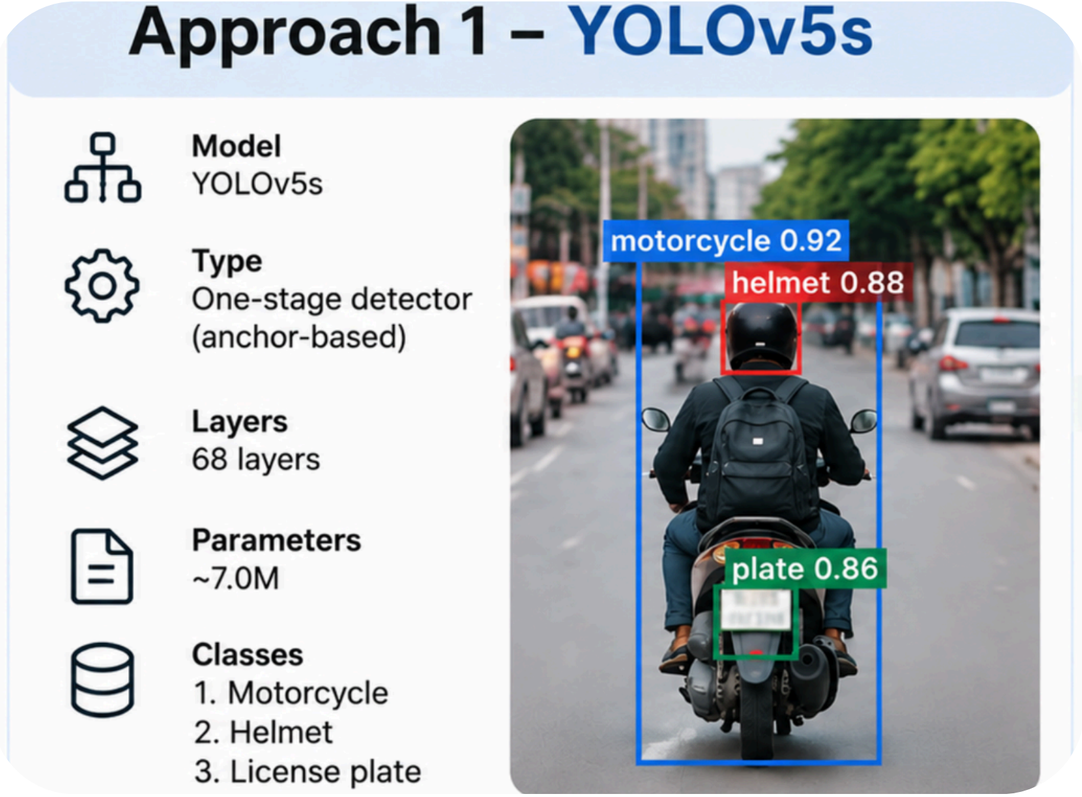
YOLOv5s	YOLOv8s
One-stage object detector	One-stage object detector
Anchor-based detection	Anchor-free detection
~7.0M parameters	~11.1M parameters
68 layers	72 layers
Pretrained model	Pretrained model
Trained on Version 5	Trained on Version 5



Goal: Compare two architectures under the same training conditions.

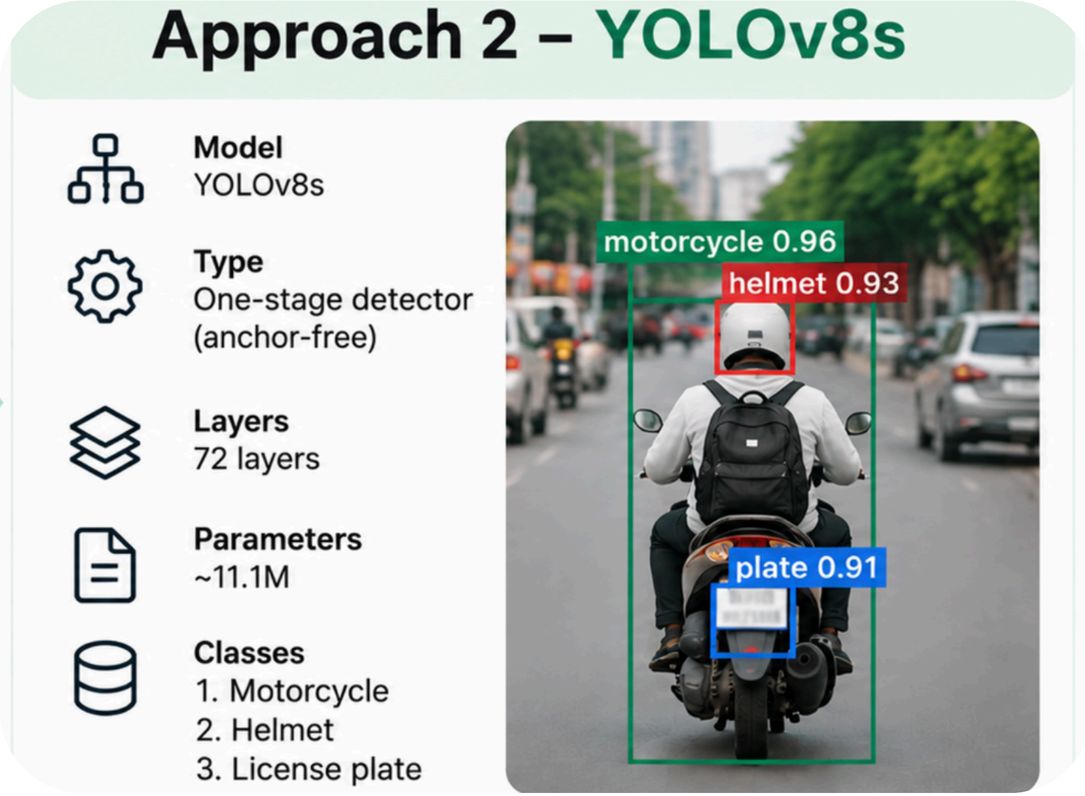
Approach 1: YOLOv5s

- One-stage object detector
- Anchor-based
- 68 layers
- ~7.0M parameters
- Trained on Version 5
- Detects your project classes: bike, helmet, no-helmet, number-plate



Approach 2: YOLOv8s

- One-stage object detector
- Anchor-free
- 72 layers
- ~11.1M parameters
- Trained on Version 5
- Same four detection classes



Experimental Setup

Fair training setup for YOLOv5s and YOLOv8s

640 × 640	16	40	Tesla T4
Image size	Batch size	Epochs	GPU

Dataset & Models

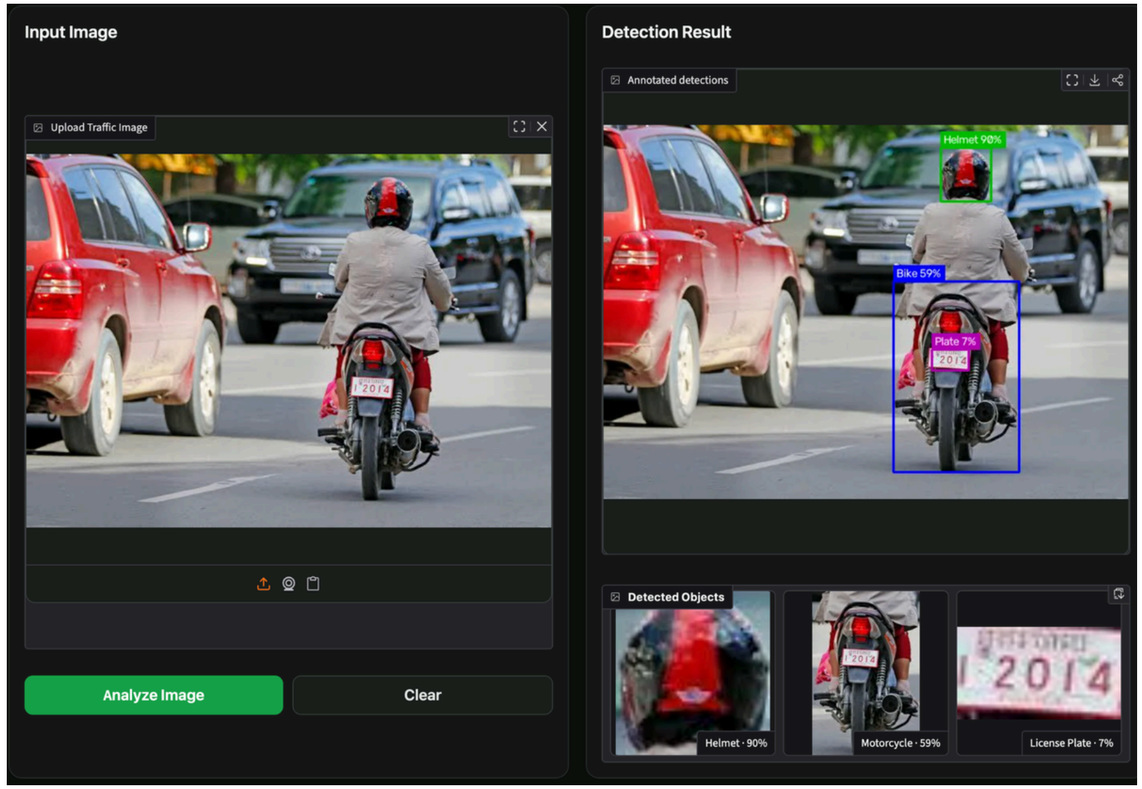
- Version 5: 8,494 images
- Train: 5,949 images
- Validation: 1,690 images
- Test: 855 images
- Models: YOLOv5s & YOLOv8s
- Framework: PyTorch + Ultralytics

Evaluation Metrics

- Precision • Recall
- mAP@50
- mAP@50:95
- No-Helmet Recall
- No-Helmet mAP@50:95

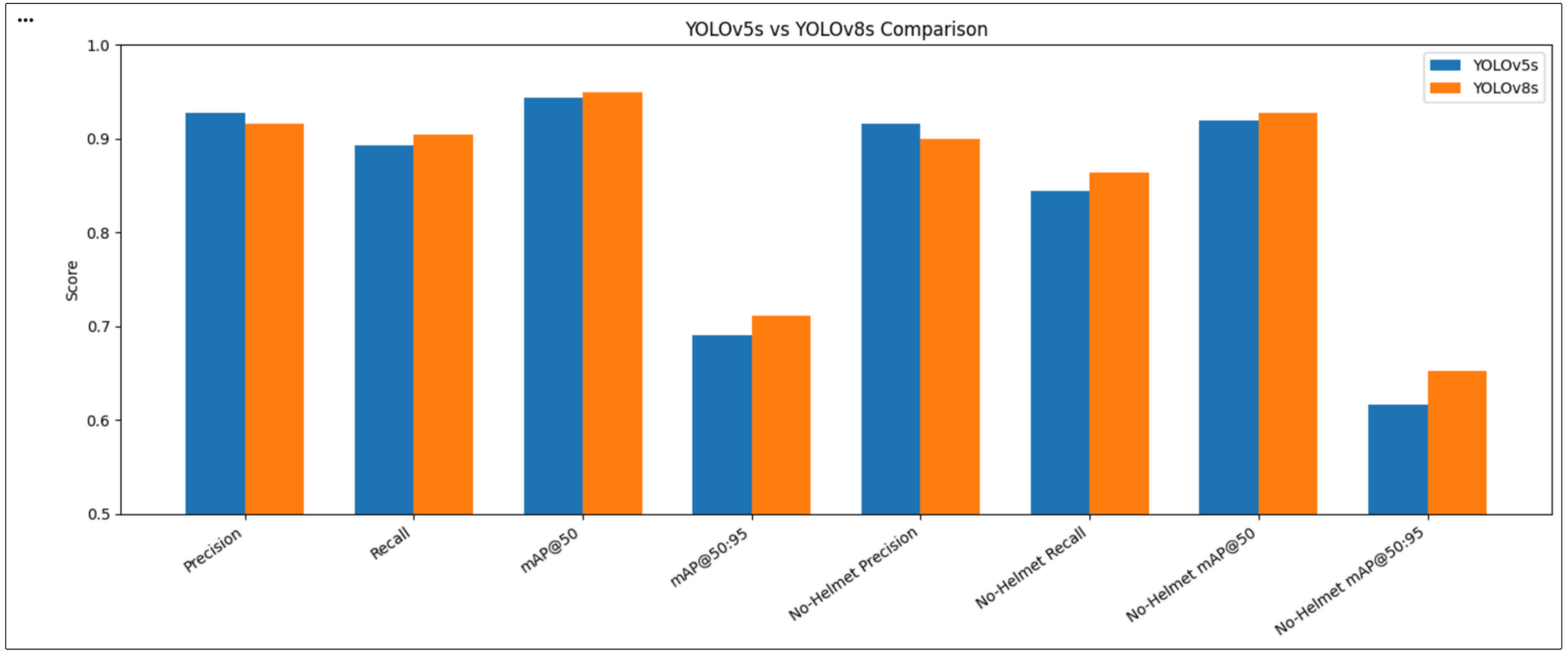
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RESULTS & VISUAL COMPARISON

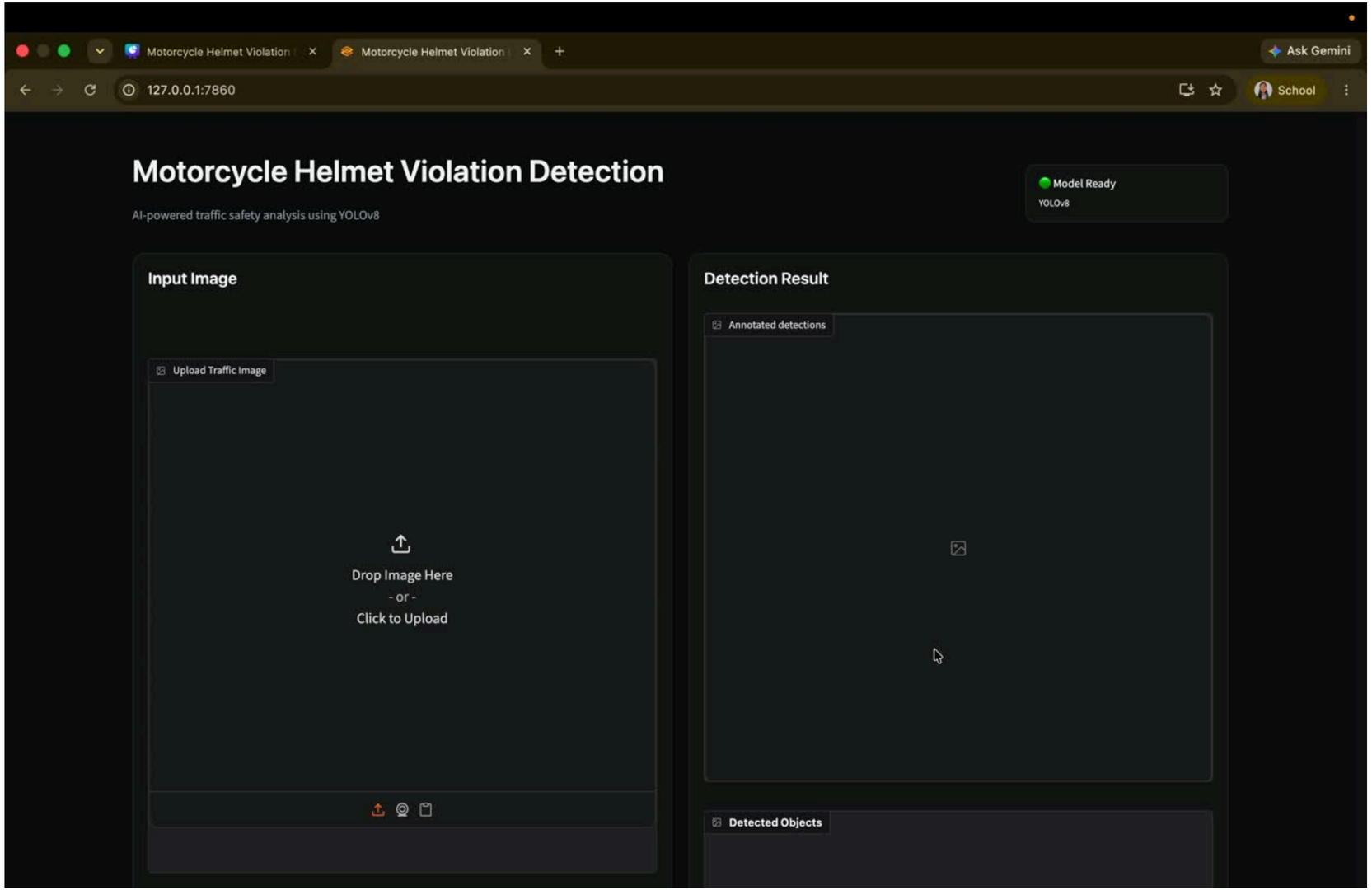


	Metric	YOLOv5s	YOLOv8s	Better
0	Precision	0.927	0.916	YOLOv5s
1	Recall	0.893	0.904	YOLOv8s
2	mAP@50	0.944	0.949	YOLOv8s
3	mAP@50:95	0.691	0.711	YOLOv8s
4	No-Helmet Precision	0.916	0.900	YOLOv5s
5	No-Helmet Recall	0.844	0.864	YOLOv8s
6	No-Helmet mAP@50	0.919	0.928	YOLOv8s
7	No-Helmet mAP@50:95	0.617	0.653	YOLOv8s

RESULTS & VISUAL COMPARISON



Demo


Motorcycle Helmet Violation
Motorcycle Helmet Violation
Ask Gemini
127.0.0.1:7860
School
Motorcycle Helmet Violation Detection
AI-powered traffic safety analysis using YOLOv8
Model Ready
YOLOv8
Input Image
Upload Traffic Image
Drop Image Here
- or -
Click to Upload
Detection Result
Annotated detections
Detected Objects

Discussion & Error Analysis

Where the model still makes mistakes and what the results show

Common Errors

- Hats or caps can be confused with helmets
- Small or distant heads may be missed
- Occlusion and rear views reduce detection quality
- Multiple riders on one motorcycle are difficult

What the Results Show

- No-helmet recall improved after fine-tuning
- Difficult no-helmet cases are detected better
- Precision decreased slightly, so false detections remain
- Crowded scenes still cause missed or incorrect detections

Conclusion & Future Work

What was achieved and what comes next

Conclusion

- Built a YOLOv8 system for helmet / no-helmet detection
- Fine-tuning improved difficult no-helmet cases
- Created a Gradio app for clear detection results

Future Work

- Add more difficult helmet / no-helmet training data
- Improve crowded scenes, multi-rider detection, bike and plate features
- Compare another detector and add video / live-camera support

References

1. Ultralytics YOLOv5: https://github.com/ultralytics/yolov5

2. Ultralytics YOLOv8: https://docs.ultralytics.com/models/yolov8/

3. Roboflow Helmet Dataset: Helmet and Number Plate Detection for Motorbike Safety

4. Local Phnom Penh Dataset

   - Koh Pich, Phnom Penh: https://youtu.be/dVp_nmBBjnk?si=0ZMLCB6lt6rNRHA9
   - Toul Tompoun, Phnom Penh: https://youtu.be/tCAWfYIPyuw?si=Rn5bWCPgWwHVv1V6

5. EasyOCR: https://github.com/JaidedAI/EasyOCR GitHub

Thank you